

DOCUMENT 2 — Stage 1 Tutorial

Activity: Why Software Engineering Still Matters

Activity 1 — Think-Pair-Share

Question: If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Software engineers still need to understand the client's requirements and what problem the software needs to solve.
2. They need to test the code and identify errors, security issues and incorrect assumptions.
3. They need to understand the code well enough to explain, maintain and improve it.

These answers address the tutorial's focus on why software engineering involves more than simply producing code.

Activity 2 — Is This Software Engineering?

Scenario	Programming ?	Software Engineering?	Why?
A — A student writes a 50-line Python calculator	Yes	Not necessarily	May not go through all engineering steps.
B — A team develops a payroll system used by 5,000 employees	Yes	Yes	Has to go through all engineering steps of Requirements design development testing maintenance.
C — AI generates a simple appointment application from one prompt	Yes	Not by itself	Requirements, assumptions and output still need to be checked and tested.

Activity 3 — SmartCare Problem Analysis

Task 1 — Identify Stakeholders

Stakeholder	What do they need?
Receptionists / Clinic Staff	Way to record, find and manage patient and appointment information
Patients	Appointment and patient information
Practitioners	Appointment information for the patients they see
Clinic Management	System that improves the clinic's current processes

Task 2 — Identify Current Problems

1. Patient and appointment information is spread across spreadsheets and paper records.
2. Information may be difficult or slow for staff to find.
3. Manual records can make information harder to keep consistent and up to date.
4. The current process may make appointment management less efficient.

Task 3 — Ask Client Questions

1. What patient information needs to be stored?
2. What information needs to be recorded for each appointment?
3. Who should be allowed to create, change or cancel appointments?
4. Should the system prevent two appointments from being booked at the same time?
5. How should staff search for patients and appointments?

The SmartCare scenario in this tutorial only states that the clinic currently uses spreadsheets and paper records and wants software to improve those processes, so these questions help clarify the missing requirements.

Activity 4 — Critique an AI Response

AI Suggestion	Client Evidence?	In Scope?	Decision
Appointment management	Yes	Yes	Accept — Directly mentioned
Facial recognition login	No	No	Reject — No client requirement
AI diagnosis recommendations	No	No	Reject — Not part of the stated problem

Patient search	Not confirmed	Possible	Keep provisional — Should be confirmed with the client
Online payment	No	No	Reject — Not mentioned
Practitioner schedule view	Not confirmed	Possible	Keep provisional — The client has not requested it
Insurance processing	No	No	Reject — Not mentioned
Treatment-plan generation	No	No	Reject — Outside the stated problem

Exit Question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

A software engineer must verify the software meets with clients but doing a checklist of requirements and see what the software does and determine if the answer is appropriate.